Laboratory 09

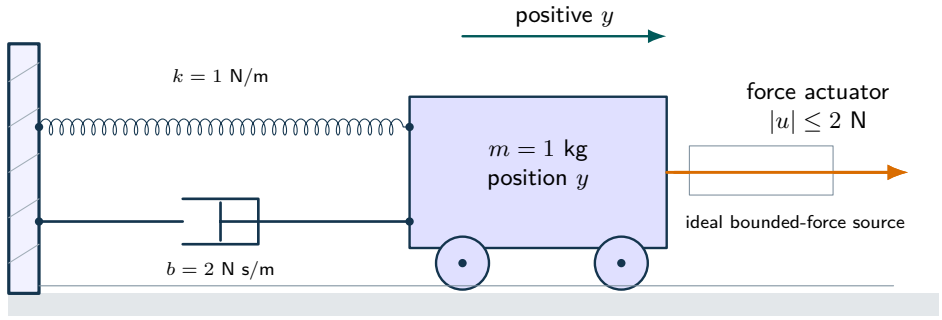
PID regulation when the actuator saturates

Exercise sheet – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal plant: $\ddot{y} + 2\dot{y} + y = u$. The physical realization has $m = 1$ kg, $b = 2$ N s/m and $k = 1$ N/m. Ideal PID pole targets are $-2, -3, -4$; constrained cases impose $|u| \leq 2$.

1. Pole placement for a force-limited positioning stage

The physical illustration gives one realization of $\ddot{y} + 2\dot{y} + y = u$: a 1 kg carriage, a 2 N s/m damper and a 1 N/m spring, driven by an ideal force actuator. Position is in m and force in N.



Illustrative force-driven stage; no contact, friction or hard stops

Force and displacement

u: applied force [N].
y: displacement [m].
Requested and applied inputs differ when saturation is active.

Dynamics and bound

$\ddot{y} + 2\dot{y} + y = u$.
$-2 \leq u \leq 2$ N.
The drawn source applies force to the moving mass.

Controller implication

Anti-windup must use applied-minus-requested effort. Wheels depict ideal guidance; rolling friction and hard stops are omitted.

A mass, spring, damper and bounded force actuator physically realize the normalized second-order plant. Force saturation is included; contact, dry friction and travel stops are not.

(a) For $G = 1/(s^2 + 2s + 1)$, use ideal PID $C = k_p + k_i/s + k_d s$. Match the closed-loop denominator to $(s + 2)(s + 3)(s + 4)$ and find all three gains.

(b) Compare $u = k_p e + k_i \int e\,dt + k_d \dot{e}$ with $u = k_p e + k_i \int e\,dt - k_d \dot{y}$, where $e = r - y$. Derive both reference numerators for the preceding plant and explain derivative kick.

(c) For a 3 m reference step, determine the impulsive contribution requested by ideal derivative-on-error action with the designed k_d. Explain why matching the nominal poles does not make that command physically realizable.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

2. Implementing a filtered controller as a state model

Replace the ideal derivative by a first-order filter and retain the resulting additional dynamics. Compare a transfer-function calculation with a state-equation calculation.

(a) Use derivative-on-measurement $D_f = 7s/(0.03s + 1)$. Derive a first-order filter-state implementation and its high-frequency gain. Give the units of the filter time constant.

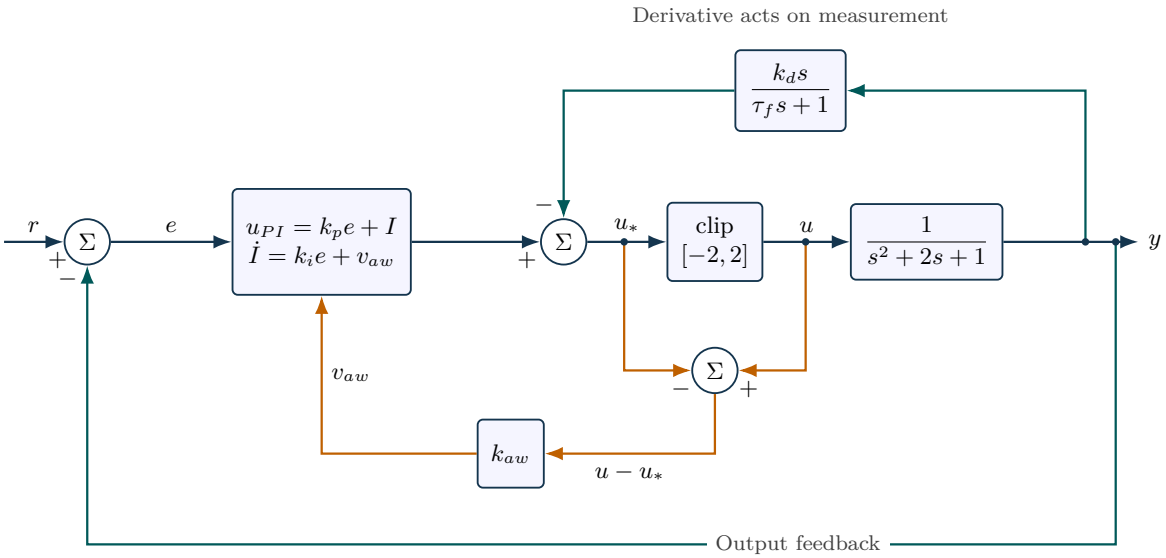
(b) With gains 25, 24 and 7 and filter constant 0.03, derive the unsaturated characteristic polynomial. Verify strict stability using the quartic Routh inequalities.

(c) Use states $[y, v, z, I]^T$, where $v = \dot{y}$ and $\tau_f \dot{z} = y - z$. Derive the unsaturated zero-reference state matrix and explain how its eigenvalues check the filtered characteristic polynomial.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

3. An unreachable reference with a bounded integral state

The requested position exceeds the equilibrium capability of the force actuator. Analyze both ordinary integration and back-calculation without confusing bounded controller state with successful tracking.



Filtered derivative acts on measurement. Back-calculation sends the actual-minus-requested actuator effort into the integral-state equation.

(a) The plant satisfies $\ddot{y} + 2\dot{y} + y = u$ and the actuator is limited to $[-2, 2]$. Is a constant reference of 3 feasible? What is the largest positive equilibrium output, and what happens to an ordinary integral state if the tracking error stays at one?

(b) Use $\dot{I} = 24e + k_{aw}(u - u_*)$ with $k_{aw} = 8$. At an instant when $e = 1$, requested effort $u_* = 5$ and saturated effort $u = 2$, evaluate $\dot{I}$. Find the request excess that balances integration for a constant positive error e.

(c) Find a saturated steady state for $r = 3$, $u_{\max} = 2$ and $k_{aw} = 8$, including y, v, z, I, u_*, for the filtered derivative-on-measurement implementation. State the remaining tracking error.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

4. Anti-windup speed and numerical step-size limits

During a fixed saturation interval, the anti-windup correction can be faster than the plant. Examine that local time scale before choosing a numerical integrator.

(a) Assume, only for this local calculation, that $e = 0.4$ and $u - u_* = -0.5$ remain constant for 2 s. Find the integral-state increment with and without back-calculation gain 8. Explain the scope of this calculation.

(b) Hold y, r and the filtered derivative contribution fixed while the actuator remains at a fixed saturation limit. Derive the homogeneous dynamics of a perturbation in integral state I. Compare its time constant for gains 8 and 40.

(c) Apply explicit Euler to the homogeneous local integral-state perturbation. Derive the strict step-size condition for its decay and evaluate it for gains 8 and 40. Explain why satisfying this condition alone is insufficient for the full simulation.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

5. A reproducible recovery comparison with an explicit evidence limit

The reference becomes feasible after an initial saturated interval. Compare recovery using both tracking and actuator records, keeping all data explicitly synthetic.

(a) A synthetic experiment commands 3 until 6 s, then 0.2 until 12 s. Define recovery IAE on the second interval. Given values 9.756216 without back-calculation and 1.841207 with it, calculate the percentage reduction and discuss what the metric omits.

(b) Specify a numerical protocol for comparing the two anti-windup settings fairly. Include the reference discontinuity, time resolution, metrics and the evidence label. Explain why plotting only the requested effort is insufficient.

(c) Suppose a design objective is $J = \mathrm{IAE} + \lambda \int_6^{12} \dot{u}(t)^2 \, dt$ with $\lambda > 0$. Can the supplied two IAE values alone rank the controllers under this objective? Specify the additional records and numerical care required.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.